

OptiMax i

PICAN devices

The PICAN devices are:

- prewinders. See PICAN devices > Prewinders [90];
- PFL;
- thermal cutters.

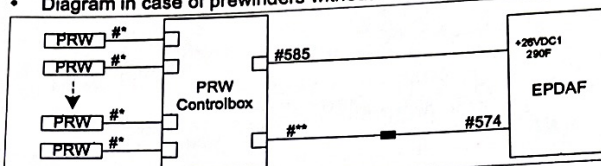
PICAN devices > Prewinders

Installation

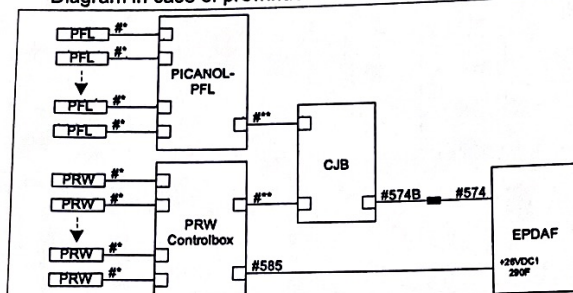
See the 'Installation' section.

Block diagram

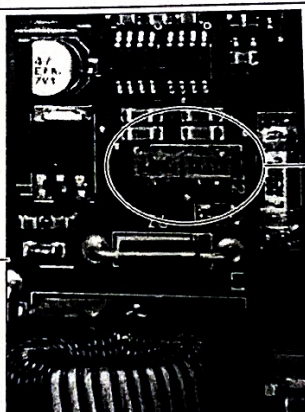
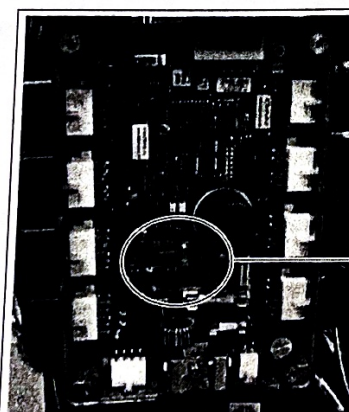
- Diagram in case of prewinders without PICANOL-PFL.



- Diagram in case of prewinders with PICANOL-PFL.



Termination on the IRO controlbox



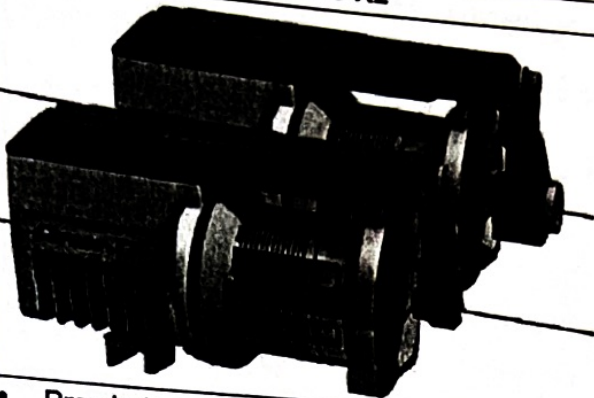
☒ Terminated	J1/J2 closed
☐ Not Terminated	J1/J2 open

- Cabling

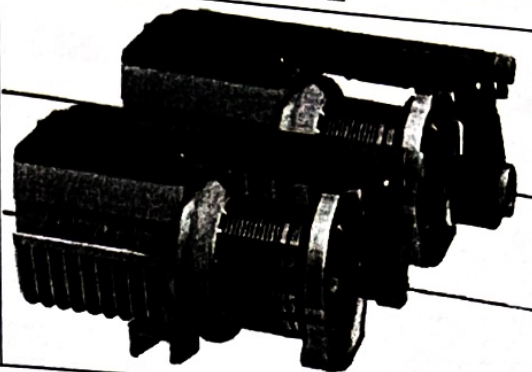
Cable (BE number)	From	To
IRO-CAN cable (BE307082)	prewinders	prewinder control box
IRO-CAN cable (BE218441)	prewinder control box	#574 or CJB

- Prewinders

Chrono X2

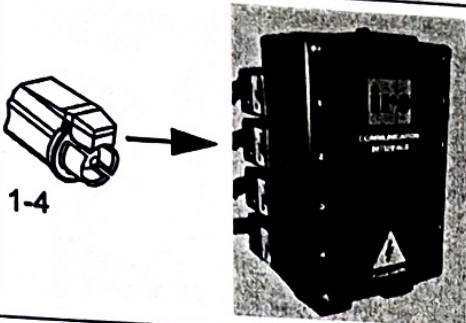


Luna X2

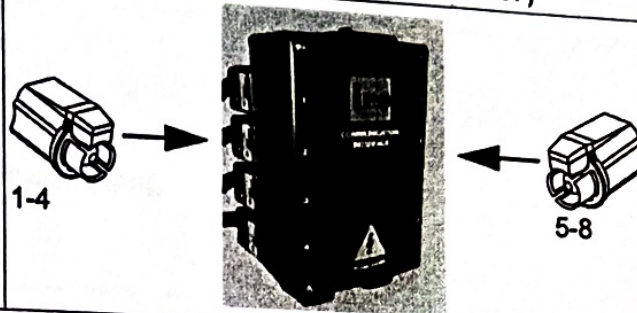


- Prewinder control boxes. There are two types of prewinder control boxes.

Up to 4 channels (BE218285)



Over 4 channels (BE218287)

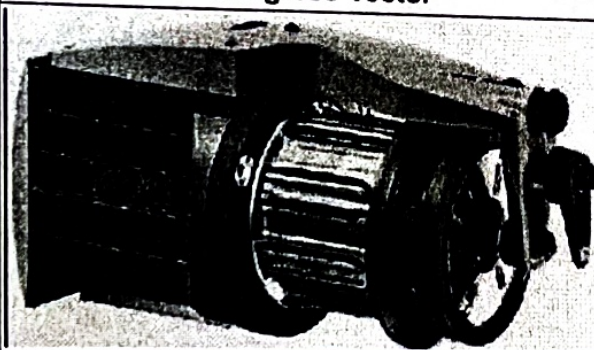


- Cabling

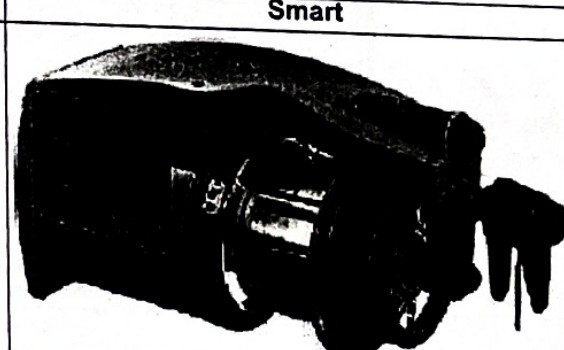
Cable	From	To
LGL-CAN cable	prewinders	prewinder control box
LGL-CAN cable	prewinder control box	#574 or CJB

- Prewinders

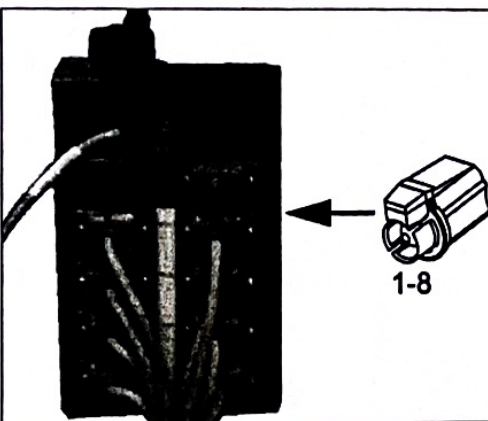
Progress Vector



Smart



- Prewinder control box



OptiMax-i

PICAN devices > PFL

General

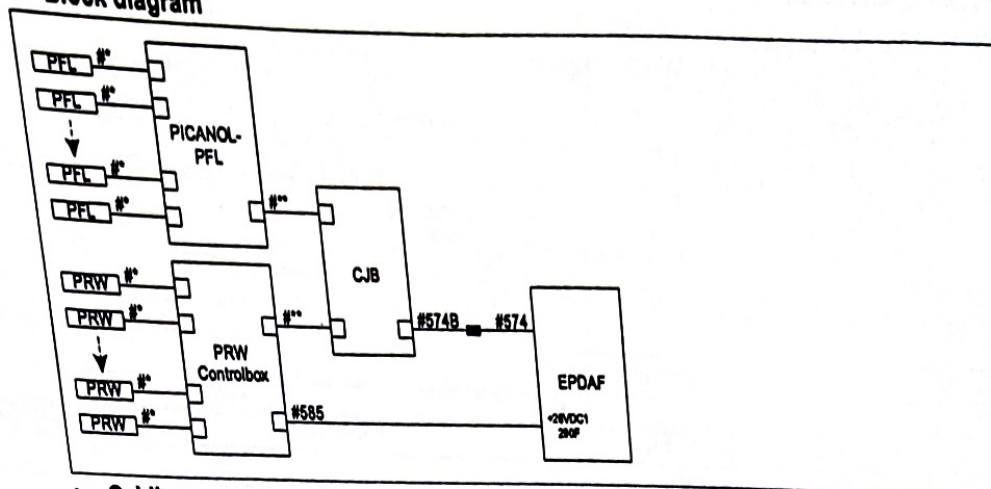
There are 3 types of PFL:

- PICANOL-PFL [93];
- IRO-PFL with central cleaning [94];
- IRO-PFL with individual cleaning [95].

Installation

See the 'Installation' section.

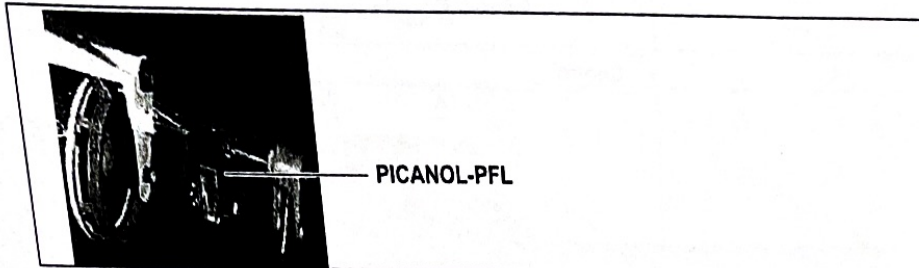
- The PFL is controlled by a separate controlbox.
- Block diagram



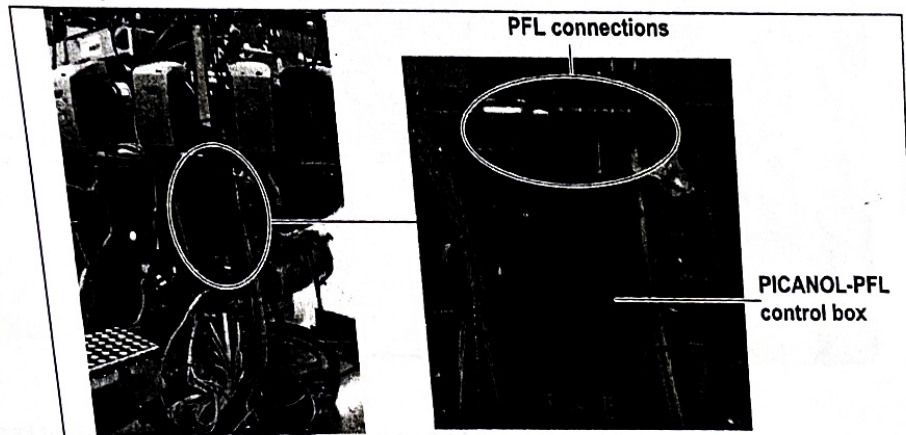
- Cabling

Cable	From	To
CAN cable	PICANOL-PFL control box	CJB
PICANOL-PFL	PICANOL-PFL control box	PICANOL-PFL

- PFL

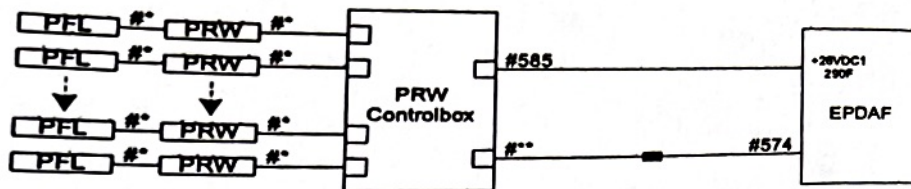


- Control



- The PFL control is controlled by the prewinder and the central cleaning valve is in the PRW controlbox.

Block diagram



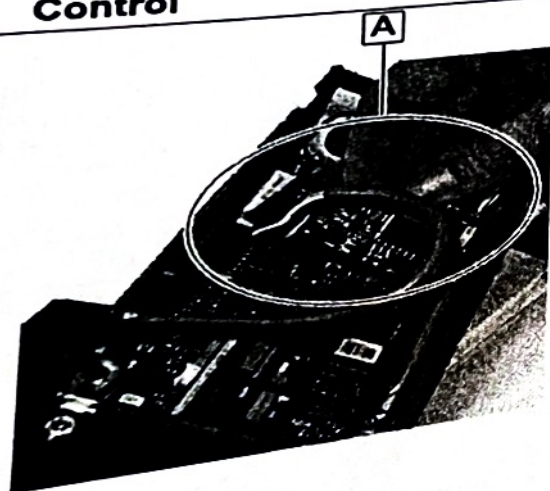
Cabling

Cable	From	To
IRO-PFL	prewinders	PICANOL-PFL

PFL



Control



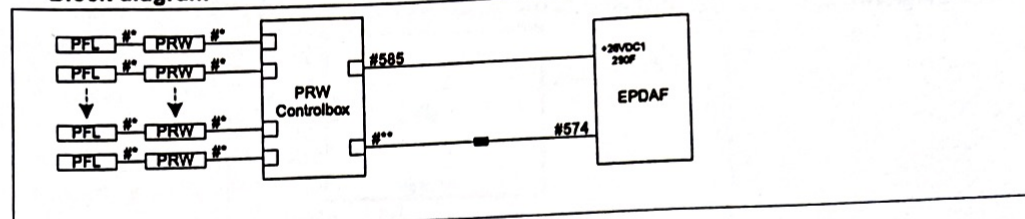
- A. IRO-PFL print
- B. Cleaning valve



In every prewinder, there is a IRO-PFL print. The cleaning valve is located in the IRO controlbox.

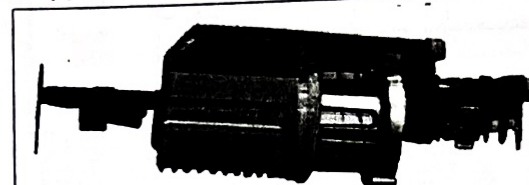
IRO-PFL with individual cleaning

- The PFL control is controlled by the prewinder and there is a cleaning valve in each prewinder.
- Block diagram**

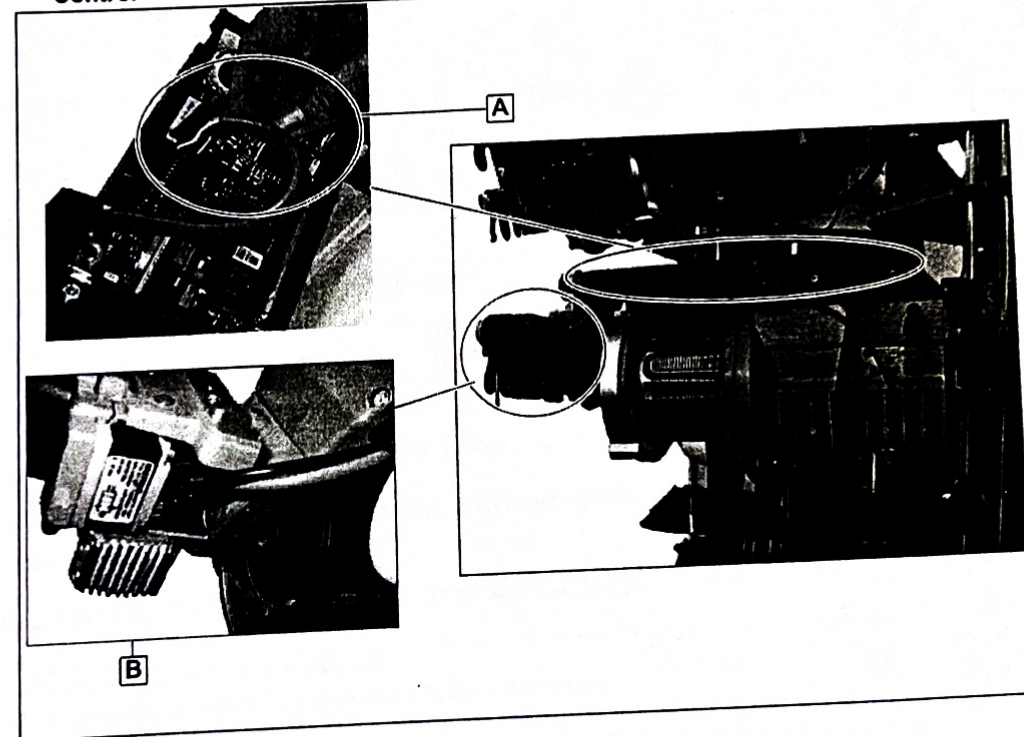


Cabling	From	To
IRO-PFL	prewinders	PICANOL-PFL

- PFL**



- Control**



A. IRO-PFL print
B. Cleaning valve



In every prewinder, there is a cleaning valve and an IRO-PFL print.

PICAN devices > Thermal cutter (eCTTH)

General

The thermal cutter module is fitted on the high voltage plate.
See [High voltage plate](#) [31].

If problems occur

Return the module as a whole to Picanol.

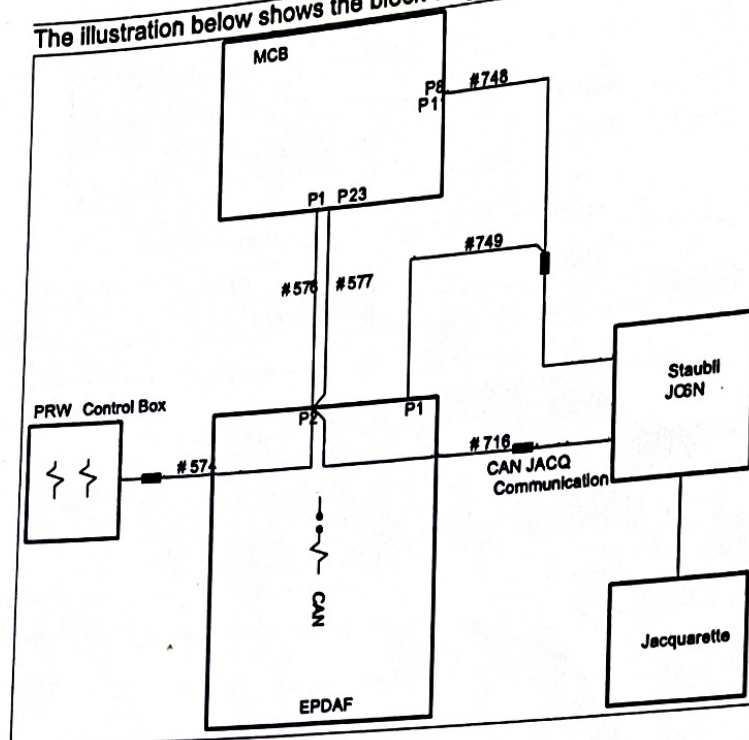
Diagram

See [Signals and cable](#) [87].
See Thermal cutter (eCTTH) > With 1 centre thermal cutter.

PICAN devices > Jacquarette

Block diagram

The illustration below shows the block diagram of the jacquarette:



Properties of the electrical components located outside the control boxes

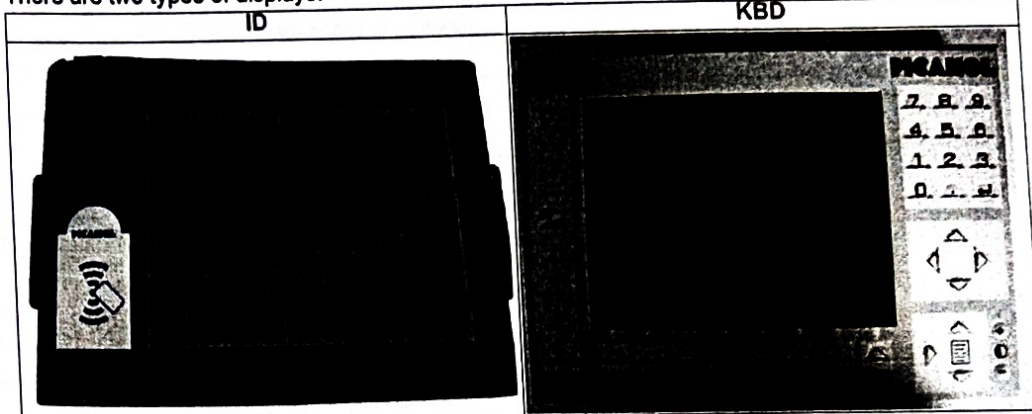
Contents

- Display characteristics [97]
- Properties of the proximity switches and other detection elements [97]
- Properties of the stepping motors [98]
- Properties of the servomotors [99]
- Properties of the valves [99]
- Properties of the safety brake [99]
- Properties of the resolver [99]
- Properties of the resolver [99]

Display characteristics

General

There are two types of displays:



Cabling

The connection is the same for both displays.

Cable	From	To
#101B	MCB (via #101A)	X-TERM
#102	X-COMP; PDGN and MCB	

Remarks



Disconnect the main power supply before (dis)connecting the display.



Make sure the support of the display is always earthed to the machine.

Properties of the proximity switches and other detection elements

Proximity switches

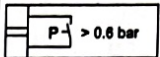
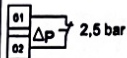
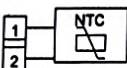
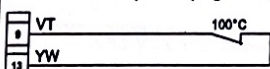
All proximity switches use 26V DC.

The proximity switches are described in the table below.

PX	Description/property
PX-WP	Detects whether the pickfinding mechanism is in the weaving position.
PX-PFP	Detects whether the pickfinding mechanism is in the pickfinding position.
PX-REF	Detects whether the sley is in the reference zone.
PX-FA-SFTY	Only with fancy beam. Signals if the back rest roller is within its functional working area.
PX-LVL-OFF	Only for outside cam motion with leveling. Returns a signal if the weaving machine is ready for weaving (Leveling OFF).

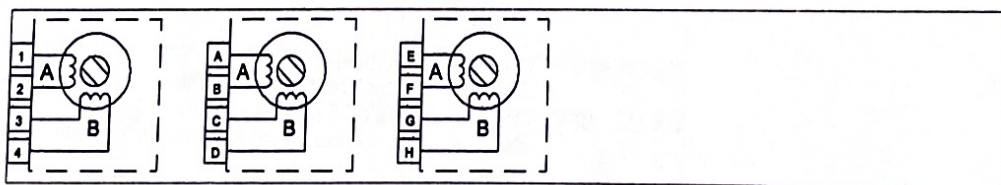
Other detection
elements

The following table describes the other detection elements.

Detection element	Description/property
DT-LU-PRESS	Pressure switch. A contact that is open when the oil pump motor is not running and no oil pressure is present as a result. There is a delay of a few seconds after the switching on of the oil pump motor before the oil pressure can be assessed. If no oil pressure is found after a few seconds, a machine stop is generated. Switching pressure = 0.6 bar. 
DT-LU-FL	Pressure differential switch for the filter. If the filter is dirty, the pressure differential will be too large and the contact opens. Switching pressure difference = 2.5 bar. 
DT-Q-M-ASP	Control signal of the 3-phase motor safety.
DT-Q-M-LVL	When one phase becomes overloaded or is interrupted, the power supply to the motor is switched off (as a safety measure). The contact opens if the protection has been activated. You must reset this protection manually.
BM-STA	Batching motion control switch. The contact is closed if the batching motion is not ready for weaving (rule of exception).
CT-TH-STA	Thermal cutter control switch.
DT-TH-LU	Measures the temperature of the oil circuit by means of an NTC (Negative Temperature Coefficient) resistor. The temperature value can be read from the machine terminal. Switching temperature = 100 °C. 
DT-TH-M-ASP	Monitors motor temperature using a bimetal or PTC (Positive Temperature Coefficient) resistor.
DT-TH-M-ELO	
DT-TH-M-ETU	The contact opens (high-ohmic) at 100 °C.
DT-TH-M-FA	
DT-TH-M-LU	
DT-LU-SH-LV	Detects the outside cam motion or dobbie oil level.

Properties of the stepping motors

Diagram

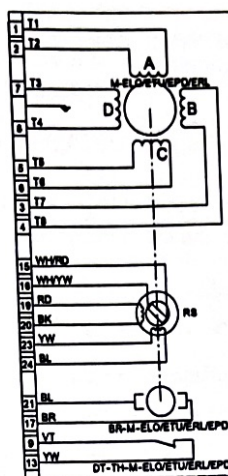


Properties

Motor	Description	Property
M-ELSY (LHS, MID1, MID2 and RHS)	ELSY device	$R_{A/B} = 0.37 \Omega$ $R_{A/B} = 22.4 \Omega$
M-WCT (RHS)	waste cutter	
M-FBR-CT (1 to 3)	fabric cutter	
M-QS	filling presenter	

Properties of the servomotors

Diagram



Properties

Motor	Description	Property
M-EL01	right-hand let-off	SR motor $R_{A/B/C/D} = 2.3 \Omega$ (cold)
M-EL02	left-hand let-off	
M-EL0-FA	fancy beam let-off motion	
M-ETU	take-up motion	

Properties of the valves

Properties

The following table describes the valves.

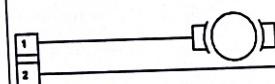
Valve	Property
pickfinder valve	$R_{V-PF} = 350 \Omega$
LHS gripper clamp cleaning valve	24V 1 W component.
RHS gripper cleaning valve	$R_{V-GRP(-CLMP)-CLN} = 576 \Omega$ (cold) - 660 Ω (hot)

Properties of the safety brake

Properties

The following table describes the brake.

Brake	Property
safety brake	$R = 30 \Omega$ at 20 °C



Properties of the resolver

Diagram

